

Lumen REST API Reference

Version: v1.0.0 | **Base URL:** http://127.0.0.1:8000/api | **Content-Type:** application/json

Lumen provides a robust, thin JSON REST API layer over its core observability services. All responses use standardized envelopes ({ "data": ... } or { "error": ... }).

Endpoint Summary

Method	Endpoint	Description
GET	/api/knowledge	List all knowledge base chunks.
POST	/api/knowledge	Create a new knowledge chunk.
PUT	/api/knowledge/{id}	Update an existing knowledge chunk.
DELETE	/api/knowledge/{id}	Delete a knowledge chunk.
POST	/api/console/query	Execute a test query against the RAG system and retrieve diagnostics.
GET	/api/messages/{id}/replay	Retrieve the trace timeline and diagnostics for a specific message.
POST	/api/messages/{id}/replay	Run a comparative replay for a historical message trace.
POST	/api/messages/{id}/evaluation	Submit a human review and evaluation signal for a trace.
GET	/api/health-score	Retrieve the platform's aggregate health score and metrics.
GET	/api/doctor/cases	Retrieve paginated diagnostic anomaly cases for review.
GET	/api/knowledge-gaps	Retrieve a ranked list of unmatched query terms (missing context).

1. Knowledge Base CRUD

GET /api/knowledge (List Chunks)

- **Returns:** Array of vectorized document chunks acting as ground-truth for RAG.

POST /api/knowledge (Create Chunk)

- **Body:** {"title": "Policy", "content": "Our policy is...",

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"tags": ["policy"]}
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- **Validation:** title and content are required strings.
- **Returns:** 201 Created with the new chunk resource.

PUT /api/knowledge/{id} (Update Chunk)

- **Body:** {"title": "Updated Title", "content": "Updated content..."}
- **Returns:** 200 OK with the updated chunk resource.

DELETE /api/knowledge/{id} (Delete Chunk)

- **Returns:** 200 OK with {"data": {"success": true}}. Removes context from the vector store.

2. Diagnostics & RAG Execution

POST /api/console/query (Execute Test Query)

- **Description:** Simulates an LLM completion, fetches context, runs diagnostics, and returns health metrics.
- **Body:** {"prompt_version_id": 2, "query": "How do I reset?", "provider": "mock"}
- **Validation:** prompt_version_id, query, and provider are required.
- **Returns:** AI response text and diagnostic metrics (groundedness, root_cause).

3. Replay & Prompt Comparative Analysis

GET /api/messages/{id}/replay (Show Replay Timeline)

- **Returns:** Complete trace timeline including context chunks, prompt version, AI output, and diagnostic scoring for a historical message.

POST /api/messages/{id}/replay (Comparative Replay)

- **Description:** Re-runs a historical failed query against a different prompt_version_id to compare outputs.
- **Body:** {"prompt_version_id": 3}
- **Returns:** The newly generated response and updated diagnostics without overwriting the original trace.

4. Human Audit & Evaluation

POST /api/messages/{id}/evaluation (Submit Evaluation)

- **Description:** Attaches a human reviewer rating and flag to a trace to assist in dataset alignment.
 - **Body:** {"reviewer_name": "Auditor", "rating": 1, "flag": "hallucination", "notes": "Fabricated."}
 - **Validation:** reviewer_name, rating, and flag are required.
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5. Analytics & Aggregation

GET /api/health-score (Retrieve Health Score)

- **Query Params:** lookback_days (Optional, defaults to 7).
- **Returns:** Global platform anomaly metrics and composite health score (score, total_queries, breakdown).

GET /api/doctor/cases (Retrieve Doctor Cases)

- **Returns:** Paginated trace logs flagged with anomalies (e.g., knowledge_gap, hallucination) for triage.

GET /api/knowledge-gaps (Retrieve Knowledge Gaps)

- **Returns:** Ranked frequency list of search keywords that historically returned 0 matching context chunks.
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Error Codes

Code	Type	Description
400	BAD_REQUEST	Malformed request or missing configuration.
404	NOT_FOUND	Requested resource ID was not found.
422	VALIDATION_ERROR	Request payload failed semantic or type validation.
500	INTERNAL_SERVER_ERROR	Critical unhandled system failure.

(Note: JWT Authentication, Pagination, RBAC, and Webhooks are slated for future releases.)